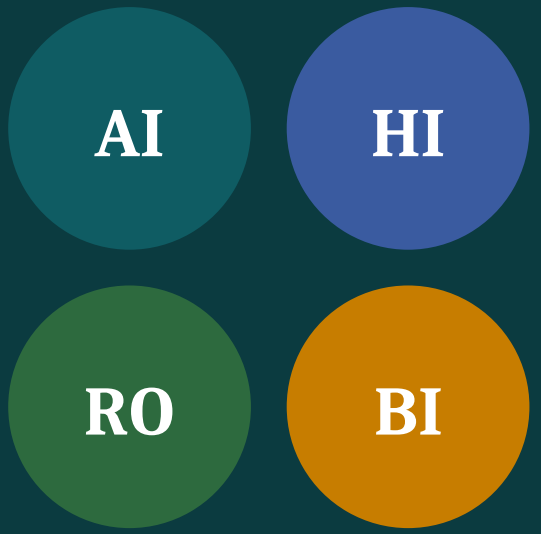
ICARB-D ANNUAL SURVEY OF DENTAL SCHOOLS

Mapping the digital maturity of dental schools

Artificial intelligence · Health informatics · Robotics · Bioinformatics

A guide for teachers



Questionnaire version 2.1.0

The question the survey answers

Which dental schools in the United States are best prepared to adopt and lead innovation in health informatics, artificial intelligence, robotics and bioinformatics, and how do they compare with dental schools in other countries?

A form can describe one of three units, and the person closest to it answers:

1 Whole dental school (school-wide)

Who answers

Dean or associate dean (academic affairs, research, clinical affairs or informatics)

2 One department

Who answers

Department chair or a faculty member designated by the chair

3 One course taught in a department

Who answers

Course director or coordinator

Why map maturity now

1

AI has reached the clinic

AI reads radiographs, digital workflows replace analogue ones, and students use generative AI every day.

2

Schools move at different speeds

Investment, training and rules vary widely, and no shared measure says who is ready or where the gaps are.

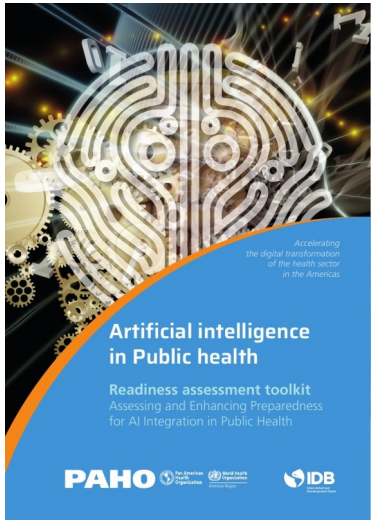
3

Evidence to plan

A common, yearly measure lets deans and faculty plan, learn from the best prepared, and see progress.

Inspired by four PAHO publications

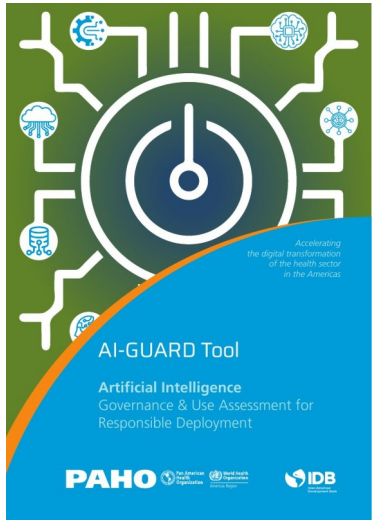
The Pan American Health Organization built a questionnaire to measure how ready public health institutions are to use AI. This survey adapts it to dental schools. Click a cover to open it on PAHO's website.



AI in Public Health: Readiness Toolkit

PAHO and IDB · 2024

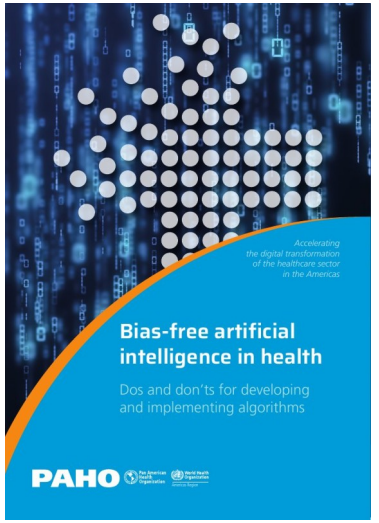
Main inspiration: the dimensions, the maturity scale and the analysis steps



AI-GUARD Tool

PAHO and IDB · 2026

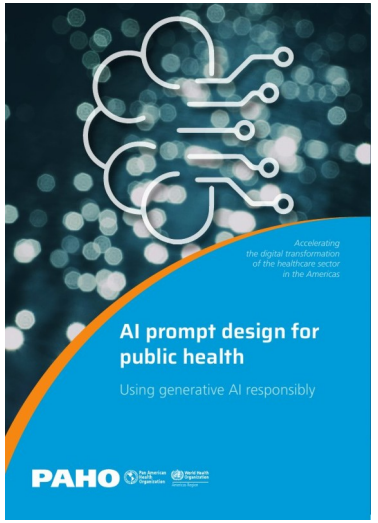
Governance checklist for AI tools (items F06 and J11)



Bias-free AI in Health

PAHO · 2026

Data representativeness and bias teaching (items C15 and J06)



AI Prompt Design for Public Health

PAHO · 2025

Prompt design and checking AI outputs (items F04–F05)

Thirteen groups of questions

175 questions in 13 parts. Part A is completed with the research team; parts B to M by the respondent, in about 45–60 minutes.

Who and what · Resources · Teaching · The four fields · Rules and lasting change · Self-view

A Profile of the unit	B Context	C Physical resources	D Human resources
E AI curriculum	F Generative AI	G Health informatics	H Robotics and digital dentistry
I Bioinformatics and omics	J Governance and ethics	K Implementation and sustainability	L Collaboration
M Self-assessment			

Part A · Profile of the unit

A

17
questions

Completed with the research team

Who and what

GOAL

Identify exactly which unit is measured and who answers for it, so every score can be traced, compared and followed from year to year.

What the questions cover

- Country, state and school
- The unit: whole school, one department or one course
- Training level and the respondent's role
- Survey year, consent and last year's code

Part B · Context

B

11
questions

Answered by the respondent

Who and what

GOAL

Measure the size of the unit, so every count becomes a fair ratio (per student, per faculty member, per year).

What the questions cover

- Students and faculty (headcount and full-time equivalent)
- Academic calendar and terms per year
- Course hours and the technology fields the unit touches

Part C · Physical resources

C

22
questions

Answered by the respondent

Resources

GOAL

Know what the unit has today: computing, software, AI tools, data, privacy and security.

What the questions cover

- Computers, GPUs, cloud access and dedicated spaces
- Software and the AI tools students actually use
- Datasets, data quality and representativeness
- Privacy, security, connectivity and IT support

Part D · Human resources

D

16
questions

Answered by the respondent

Resources

GOAL

Know who can teach, build and support these technologies, and what holds them back.

What the questions cover

- Faculty able to teach AI, and faculty doing AI research
- Training received and planned
- Partners and technical staff
- Openness to change and the main barriers

Part E · AI curriculum

E

24

questions

Answered by the respondent

Teaching

GOAL

Know what students actually learn about AI, at which level and for how many hours.

What the questions cover

- Hours per year, by teaching format and by 16 topics
- Programming and formal AI model development
- Named predoctoral and graduate courses
- Assessment, materials and planned changes

Part F · Generative AI

F

8

questions

Answered by the respondent

Teaching

GOAL

Know whether generative AI is used with clear rules, real skills and working safeguards.

What the questions cover

- Policy and permitted uses for students and faculty
- Prompt design and checking AI outputs
- Safeguards in place (adapted from PAHO AI-GUARD)

Part G · Health informatics

G

12
questions

Answered by the respondent

The four fields

GOAL

Know whether the clinic produces the reliable digital data that teaching, research and AI depend on.

What the questions cover

- Electronic records and imaging archives
- Coding, teledentistry and analytics
- Access to clinical data for research
- Teaching hours and informatics staff

Part H · Robotics and digital dentistry

H

10
questions

Answered by the respondent

The four fields

GOAL

Know what robotic and digital equipment exists, and whether students really use it.

What the questions cover

- Number of robots, simulators, scanners, CBCT, milling units and 3D printers
- How students get hands-on access
- Digital workflow, maintenance and trained faculty
- Teaching hours and research

Part I · Bioinformatics and omics

I

10
questions

Answered by the respondent

The four fields

GOAL

Know whether the unit can produce, analyse and share biological data.

What the questions cover

- Laboratories, sequencing, biobanks and computing
- Pipelines and people with bioinformatics skills
- Teaching hours and research areas
- How data are shared (FAIR principles)

Part J · Governance and ethics

J

12

questions

Answered by the respondent

Rules and lasting change

GOAL

Know whether technology is used safely, ethically and under clear responsibility.

What the questions cover

- Written plan, leadership and oversight
- Ethics and bias teaching
- Standards and guidance in use (ADA, CODA, HIPAA, FDA, NIST)
- Patient consent and a governance checklist for AI tools

Part K · Implementation and sustainability

K

14

questions

Answered by the respondent

Rules and lasting change

GOAL

Know whether projects move from pilot to routine use, are evaluated, and can last.

What the questions cover

- Pilot projects and whether they scale
- Metrics, evaluation and feedback
- Cost-benefit and sustainability
- Funding sources, budget and grants

Part L · Collaboration

L

8

questions

Answered by the respondent

Rules and lasting change

GOAL

Know whether the unit is ready to share data and work with other institutions on AI.

What the questions cover

- Interest in multi-institution AI projects
- De-identified data that could be shared
- Preferred ways to collaborate
- What stands in the way

Part M · Self-assessment

M

11

questions

Answered by the respondent

Self-view

GOAL

Compare how the unit sees itself with what its answers show, and record its most urgent needs.

What the questions cover

- Overall maturity in each of the four fields
- Main barriers and facilitators
- The single most urgent need
- Innovative practice and the vision for 2030

One scale for every field

Most items are rated from 1 to 4. "Don't know" is allowed and never counted as zero. Each field gets a 0–100 index, and the indices combine into the school ranking.

1

Absent

Nothing in place yet.

2

Initial

Interest or informal, isolated efforts.

3

Developing

In place in part, or not yet routine.

4

Established

Routine, documented and supported.

After the survey: a checklist for improvement

The same items become 88 checkpoints. For each one, the target is level 4 and the next step is level 3.

1

Answer

Complete the survey for the school, a department or a course.

2

See where you stand

Read the 0–100 index for each field and the detail behind it.

3

Choose a target

Decide the level you want to reach, field by field.

4

Tick the checklist

Mark what is already true; the rest are your next steps.

5

Measure again

Answer next year and see the progress.

The full checklist is online at igoralves1.github.io/AI-assessment-DS-2026/#/questions?go=checklist and in chapter 4 of the questionnaire (PDF).

A clear picture, every year

Every question, what it captures and how it is scored:
igoralves1.github.io/AI-assessment-DS-2026/#/questions

